NORVALE THERMAL SYSTEMS

AquaTherm X200 Industrial Water Chiller

Service Manual v2.3

Applies to models X200, X200-L and X200-HD with TCU-5 controller

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For use by qualified refrigeration and electrical technicians only. Read the safety section before carrying out any work on the unit.

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Document Control

This service manual is issued by Norvale Thermal Systems for use by refrigeration and electrical technicians who install, commission, maintain and repair AquaTherm X200 series chillers. It supersedes all earlier editions. Where a local regulation is stricter than a procedure described here, the local regulation takes precedence.

Revision history

Edition	Date	Summary of changes
v2.0	May 2023	First edition covering X200 and X200-L.
v2.1	February 2024	Added the dual-circuit X200-HD; new spare parts list.
v2.2	November 2024	Extended error code reference; controller firmware v2.2.7 described.
v2.3	March 2026	Firmware v2.3.0 and v2.3.1 behaviour; revised warranty terms; P-series codes added.

Conventions used in this manual

Three levels of hazard statement appear in the text. A WARNING identifies a hazard that could result in death or serious injury. A CAUTION identifies a hazard that could damage the equipment or the installation. A NOTE gives supplementary information that makes a task easier or explains controller behaviour.

Pressures are stated as gauge pressure in bar unless marked otherwise. Temperatures are in degrees Celsius; temperature differences are given in kelvin (K). Part numbers take the form PN-nnnn-x and should always be quoted in full when ordering, because several parts share the same four-digit root and differ only in the suffix letter.

Sensor identifiers (T1 to T6, P1 to P3, FS1) match the labels printed on the wiring harness and shown on the controller display. Maintenance tasks are identified as M-01 to M-16 and are described in Section 5.

1 Overview

1.1 About this manual

The AquaTherm X200 series comprises air-cooled, packaged water chillers intended for process cooling in plastics moulding, food processing, laser cutting, printing and light chemical plants. Each unit contains a complete vapour-compression refrigeration circuit, a brazed-plate evaporator, an air-cooled condenser with EC fans and a TCU-5 microprocessor controller. The chilled fluid is delivered to the customer's process loop through grooved water connections at the rear of the unit.

This manual concentrates on service tasks. Selection data, hydraulic module options and accessory kits are covered in the separate product catalogue. Section numbers are referenced throughout; the warranty clauses reproduced in Section 9 carry their own numbering and should not be confused with the section numbers of this manual.

1.2 Product family

Three model variants share a common chassis, controller and service philosophy:

- **X200** - the standard single-circuit chiller with one variable-speed scroll compressor, nominal capacity 62 kW at 12/7 °C water and 35 °C ambient air. Leaving water setpoint range 5 to 20 °C.
- **X200-L** - the low-temperature variant, fitted with an enlarged evaporator, an additional suction accumulator and a modified expansion valve program. It produces glycol solutions down to -8 °C leaving temperature, with a nominal capacity of 44 kW at -5 °C.
- **X200-HD** - the heavy-duty variant with two independent refrigerant circuits, two compressors, reinforced water headers and a longer condenser section. Nominal capacity is 88 kW and the unit remains in operation at ambient temperatures up to 50 °C.

Unless a paragraph names a specific variant, the information applies to all three. The standard X200 and the X200-L accept a maximum water-side working pressure of 6.5 bar; the X200-HD differs in this respect, as described in Section 10.

1.3 Principle of operation

The chiller removes heat from the process fluid in the evaporator, where low-pressure refrigerant boils at a temperature a few kelvin below the leaving fluid temperature. The compressor raises the refrigerant vapour to a pressure at which it can condense against outdoor air in the finned condenser. Liquid refrigerant then passes through a filter-drier and the electronic expansion valve (EEV), which meters flow back into the evaporator in response to the measured suction superheat.

Capacity is matched to demand by varying compressor speed between 30 and 100 percent. On the X200-HD a second compressor is staged in when the first reaches its upper speed limit for longer than the staging delay. Condenser fan speed is modulated to hold a stable head pressure, which protects the expansion valve from starving at low ambient temperatures.

Heat rejection depends on unobstructed airflow across the finned condenser. Deposits of airborne dust, pollen, oil mist or salt spray on the fin pack act as an insulating layer and at the same time restrict the passage of air. As a result, fouled condenser surfaces raise the condensing pressure and erode capacity, while the compressor draws more current to deliver less refrigeration effect. A heavily loaded fin pack can lift the condensing temperature by 8 K or more compared with a new coil, which moves the unit closer to its high-pressure cutout on warm afternoons.

The compressor motor is cooled by the returning suction gas. Anything that reduces suction gas mass flow or raises its temperature - a short refrigerant charge, excessive superheat, or a very high compression ratio - also raises motor winding temperature. The protection devices that respond to this are described in Section 6.5, and the corresponding fault codes in Section 8.

1.4 Major assemblies

Assembly	Location	Remarks
Refrigeration module	Lower front bay	Compressor(s), EEV, filter-drier, receiver
Evaporator	Lower rear bay	Brazed-plate, insulated, with drip tray below
Condenser section	Upper bay	Finned coil with two EC fans (three on X200-HD)
Electrical panel	Right-hand end	Isolator, drive, TCU-5 controller, terminal rail
Hydraulic connections	Rear panel	Grooved DN65 (DN80 on X200-HD)

1.5 Model designation and serial plate

The serial plate is riveted to the inside of the electrical panel door. It shows the model variant, the serial number, the refrigerant type and charge per circuit, the maximum allowable pressures on the high and low sides, and the electrical supply rating. The serial number encodes the year and week of manufacture in its first four digits; for example, 2547-01183 was built in week 47 of 2025.

Always quote the complete serial number when requesting technical support or ordering parts. Several components changed specification during production, and the correct item can only be identified from the build week.

2 Safety

2.1 Qualified personnel

Work on the refrigerant circuit may only be carried out by persons who hold a current certificate of competence for handling flammable refrigerants of safety class A2L. Work inside the electrical panel may only be carried out by persons qualified for work on low-voltage installations. Operators may use the controller display and acknowledge advisory messages but should not open any access panel.

2.2 Electrical hazards

WARNING Opening the main isolator does not remove every hazardous voltage from the unit. Stored energy remains in the compressor drive for some time after isolation.

The variable-speed compressor drive stores energy in its DC link after the supply has been isolated. Allow no less than five minutes for the bus capacitors to discharge, then confirm with a meter that the voltage measured between test points TP1 and TP2 has fallen below 50 V DC before the drive cover is lifted. The neon indicator on the drive fascia is not a reliable proof of a dead circuit.

The crankcase heater circuit on the X200-L is supplied from a separate fused spur so that it remains energised when the main isolator is open during winter shutdown. Isolate fuse F7 as well as the main isolator before touching the compressor terminal box on that variant.

2.3 Refrigerant handling

All X200 series units are charged with R-454B, a mildly flammable refrigerant of safety class A2L. It is heavier than air and can accumulate in pits, trenches and basements. Before any brazing, ensure the circuit has been recovered and purged with oxygen-free nitrogen and that a refrigerant detector shows no reading within 2 m of the work area.

Refrigerant escaping from a service port expands instantly and can freeze skin tissue and the cornea on contact. Before connecting a gauge manifold or recovery unit, wear chemical splash goggles and thermally insulated gloves, and keep sleeves rolled down. Work from the upwind side when outdoors, and never direct a hose outlet towards another person.

2.4 Pressure hazards

The high-pressure side of the circuit is designed for a maximum allowable pressure of 45 bar and is protected by a spring-loaded relief valve set at that value. The low-pressure side is designed for 28 bar. The electronic high-pressure cutout opens at 41.5 bar and a

separate mechanical pressure switch opens at 43.0 bar as a second line of defence.

On the water side, the standard X200 and the X200-L must not be exposed to more than 6.5 bar. If the customer's loop is filled from a mains supply, fit a pressure-reducing valve and a relief valve set no higher than 6.0 bar upstream of the chiller.

2.5 Lifting and handling

Lift the unit only by the four lifting eyes on the base frame, using a spreader bar so that the slings do not bear against the condenser coil or the side panels. Fork-lift pockets are provided on the long sides and must be used with forks at least 1.8 m long. Never lift the unit with water in the evaporator.

2.6 Lockout and tagout

Before starting any maintenance task, lock the main isolator in the OFF position with a personal padlock, attach a tag identifying the person working on the unit, and verify absence of voltage at the incoming terminals. If the process loop is to be opened, close and lock the external isolating valves and relieve water pressure through the drain cock on the evaporator outlet.

2.7 Ventilation and ignition sources

Because R-454B is mildly flammable, no source of ignition may be present within 3 m of the unit while the refrigerant circuit is open. Use only tools and leak detectors rated for A2L refrigerants. If the chiller is installed in an enclosed plant room, the room must have mechanical ventilation interlocked with a refrigerant detector as required by the applicable installation standard.

3 Installation

3.1 Receiving and inspection

Inspect the unit on arrival for transport damage before signing the delivery note. Check the tilt and shock indicators on the packaging; a triggered indicator must be recorded on the delivery note and reported to Norvale within 7 days. Confirm that the holding charge gauge fitted to the refrigeration module shows positive pressure, which indicates that the circuit is intact.

3.2 Siting and clearances

The chiller must stand outdoors or in a well-ventilated area with free air supply to the condenser. Maintain at least 1.2 m clear in front of the electrical panel and refrigeration module for service access, 1.0 m along each long side for air inlet, and 2.5 m unobstructed above the fan discharge. Where two units are placed side by side, the clearance between them must be 2.0 m to prevent one unit drawing the discharge air of the other.

Avoid sites where the unit is exposed to the exhaust of cooling towers, kitchen extracts or diesel generators. Such locations shorten the interval between condenser cleanings considerably and may invalidate the corrosion protection offered by the standard coil coating.

3.3 Foundations and anti-vibration mounts

Place the unit on a level concrete plinth capable of carrying the operating weight given in Section 10. The base frame must be level within 5 mm across its length. Fit the six rubber-in-shear mounts supplied with the unit (eight on the X200-HD), and connect the water pipework through flexible connectors so that compressor vibration is not transmitted into the building structure.

3.4 Water piping

Connect the process return to the inlet marked WATER IN and the supply to the outlet marked WATER OUT. Reversing the connections disables the counter-flow arrangement of the evaporator and reduces capacity by roughly 15 percent. A Y-type inlet strainer with 0.8 mm mesh, part number PN-4410-S, must be installed within 1.5 m upstream of the evaporator inlet; operation without it voids the evaporator warranty.

Install isolating valves, thermometer pockets and pressure gauge tappings at the inlet and outlet so that the evaporator pressure drop can be measured independently of the P3 differential transducer. Fit an automatic air vent at the highest point of the loop and a drain valve at the lowest point.

CAUTION Never pressure-test the process loop with the chiller connected at more than the maximum water-side pressure stated on the serial plate. Isolate the unit before testing.

3.5 External pump arrangement

Where the system uses an external circulating pump, locate it so that it pushes water into the evaporator rather than drawing through it. If the pressure at the pump suction drops close to the vapour pressure of the water, vapour bubbles form at the impeller eye and collapse violently as they reach higher-pressure regions, producing a characteristic rattling noise, pitting of the impeller and erratic flow readings. Maintain a static head of at least 1.5 m above the pump inlet and keep suction pipework short and generously sized.

The pump must be interlocked with the chiller so that the compressor cannot run without flow. Terminals X4:11 and X4:12 provide a volt-free pump start signal; the pump should continue to run for 120 seconds after the compressor stops, which the controller handles automatically through its pump overrun timer.

3.6 Fluid requirements for the X200-L

The X200-L is designed for leaving fluid temperatures below 0 °C and must never be operated on plain water. Fill the circuit with inhibited propylene glycol at a concentration between 30 and 40 percent by volume, premixed with demineralised water before it enters the system. Automotive coolants contain silicates that coat heat transfer surfaces and are not acceptable. Verify concentration with a refractometer after filling and at the intervals given in Section 5.

Enter the measured concentration into the controller under Settings > Fluid > Glycol %. The controller uses this value to set the freeze protection limit and to correct the capacity calculation.

3.7 Electrical connection

The unit requires a 400 V three-phase 50 Hz supply with protective earth; a neutral conductor is not required. Size the supply cable from the minimum circuit ampacity given in Section 10 and protect it with the maximum fuse rating stated on the serial plate. Voltage imbalance between phases must not exceed 2 percent. Phase rotation is checked by the controller at every power-up.

Control wiring for remote start/stop, alarm output and building management communication is routed through the separate low-voltage gland plate. Keep control cables at least 300 mm away from power cables or run them in a segregated trunking compartment.

3.8 Condensate drainage

Moisture from the surrounding air condenses on the cold evaporator shell and suction pipework and runs into the stainless drip tray below the refrigeration module. The tray empties through a 22 mm drain spigot at the rear left corner, which must be connected to a trapped drain with a continuous fall. Sludge and insect debris gradually obstruct the outlet if it is never flushed; a blocked drain spigot causes the drip tray to overflow, and the resulting leakage gathers around the base frame, where it is often mistaken for a leak in the process loop.

3.9 Commissioning checklist

- Confirm clearances, level and anti-vibration mounts as described in 3.2 and 3.3.
- Flush the process loop, then clean the inlet strainer before first start.
- Vent all air from the evaporator and verify flow with the P3 reading.
- On the X200-L, verify glycol concentration and enter it in the controller.
- Check supply voltage, phase imbalance and earth continuity.
- Energise the crankcase heater at least 12 hours before the first compressor start.
- Set the flow switch type parameter and confirm FS1 opens when the pump stops.
- Record commissioning data on form NTS-CF-200 and return it to Norvale to activate the warranty.

4 Operation

4.1 Controller and display

The TCU-5 controller is mounted on the DIN rail inside the electrical panel and is operated from the 7-inch touch display on the panel door. The home screen shows leaving and entering water temperature, setpoint, compressor speed and any active alarm. A three-level password system restricts access: operator (no password), service (level 2) and factory (level 3).

The controller keeps an alarm history of the last 200 events with time stamps, and a trend log of the principal temperatures and pressures sampled every 30 seconds for the previous 14 days. Both can be exported to a USB memory stick from the Service menu.

4.2 Start-up sequence

When a start command is received, the controller first starts the pump output and waits for FS1 to confirm flow for 20 continuous seconds. It then checks that the crankcase heater has been energised for long enough, that no lockout alarm is active and that the anti-short-cycle timer has expired. The compressor then starts at 40 percent speed and ramps towards the demanded speed at 2 percent per second. The EEV opens to a pre-positioned value before the compressor starts, to avoid a suction pressure dip.

4.3 Setpoints and control modes

The controller regulates on leaving water temperature (T3) by default. Entering water control (T1) can be selected for processes with large water volumes and slow load changes. The proportional band and integral time are factory-set for typical process loops and should only be adjusted after reviewing the trend log. A remote setpoint can be supplied as a 4-20 mA signal on terminals X5:3 and X5:4.

4.4 Compressor protection timers

Rapid start-stop sequences place heavy mechanical and electrical stress on the compressor, because each start draws inrush current and briefly starves the scroll set of oil. The TCU-5 controller therefore enforces a minimum off-period of 180 seconds between compressor starts and a minimum run period of 120 seconds once started. When the connected load is much smaller than the capacity of the unit, or the fluid volume in the loop is too small, the control band is crossed quickly and the unit spends much of its time in a timer-wait state. A buffer tank giving at least 6 litres of loop volume per kW of capacity is the usual remedy.

The controller also limits compressor starts to 10 per hour. If this limit is reached, code E-2105 is logged and the compressor is held off until the rolling one-hour window allows a further start.

4.5 Lead-lag rotation on the X200-HD

On the X200-HD the two compressors serve independent refrigerant circuits. To equalise bearing wear and running hours, the controller alternates the lead and lag compressor every 24 running hours, and also at each restart following a full stop. If one circuit is locked out by a fault, the other continues alone and the rotation is suspended until the fault is cleared.

Rotation can be disabled for testing under Service > Compressors > Rotation, but must be re-enabled before the service visit ends. Running with rotation disabled for extended periods is treated as abnormal operation for warranty purposes.

4.6 Controller firmware

The installed firmware version is shown under Info > Software. Firmware is updated from a USB memory stick with the unit stopped; the update takes about four minutes and all parameters are retained. Do not remove power during an update.

Firmware v2.2.7 was the production release from November 2024. Units on this build latch in a stopped state after a supply interruption and must be restarted from the display. Firmware v2.3.0 introduced a configurable auto-restart function; automatic restart after a mains failure requires firmware v2.3.0 or later, and is enabled under Settings > Power > Auto Restart.

Firmware v2.3.1 is a maintenance release and is the recommended version for all units. It corrects a fault in v2.3.0 in which the anti-short-cycle timer reset to zero after a brief power interruption, which could allow the compressor to restart immediately after a supply dip. It also introduces the flow switch type parameter FS.TYP, which must be set to PADDLE or THERMAL according to the switch fitted.

4.7 Night setback

Night setback raises the leaving water setpoint by a configurable offset and caps condenser fan speed at 70 percent between two programmed times. It is intended for sites where the process load drops overnight and where fan noise at the property boundary is a concern. While night setback is active, the display shows advisory code P-4012; this is not a fault.

4.8 Shutdown

For a normal stop, use the STOP key or open the remote start contact. The compressor ramps down and stops, the pump continues for its overrun period, and the unit remains in standby with the crankcase heater energised. For a seasonal shutdown of the X200 or X200-HD in freezing climates, drain the evaporator completely and blow it through with compressed air, or fill the loop with glycol.

5 Maintenance Schedule

5.1 General

The intervals in this section assume operation in a clean industrial environment for up to 6,000 hours per year. Sites with airborne dust, fibres or corrosive contaminants, or units running continuously, require shorter intervals at the discretion of the service provider. The controller raises advisory code P-1001 when the programmed service interval has elapsed; the counter is reset from the Service menu once the visit is complete.

Record every maintenance visit in the logbook supplied with the unit. Warranty claims are assessed against this record, and a missing entry for a task that would have revealed the fault is likely to lead to the claim being declined.

5.2 Summary of tasks

Task	Description	Interval
M-01	Visual check and alarm history review	Weekly
M-02	Log water temperatures and P3 reading	Weekly
M-03	Clean inlet strainer PN-4410-S	Monthly
M-04	Inspect and wash condenser fin pack	3 months
M-05	Refrigerant leak inspection	3 or 6 months
M-06	Glycol concentration check (X200-L)	6 months
M-07	Re-torque power terminals	12 months
M-08	Fan motor bearing and blade check	12 months
M-09	Functional test of FS1 flow switch	12 months
M-10	Process water analysis	12 months
M-11	Flush drip tray and drain	12 months
M-12	Replace filter-drier PN-2208-F	8,000 h
M-13	Replace controller clock battery	5 years
M-14	Comprehensive annual inspection	12 months
M-15	Compressor oil acid test	16,000 h
M-16	Replace flow switch paddle	5 years

5.3 Task descriptions

M-01 Visual check. Walk round the unit and look for oil stains, damaged insulation, loose panels and debris on the condenser face. Review the alarm history on the display and note any code that has occurred more than once since the previous check.

M-02 Temperature and pressure log. Record entering and leaving water temperatures, ambient temperature and the evaporator differential pressure shown for P3. A gradual rise in P3 at constant flow is an early sign of strainer blockage or scaling inside the evaporator plates.

M-03 Inlet strainer. Isolate the strainer, remove the screen and wash it in clean water. Replace the screen if any mesh is torn. In the first month after commissioning this task may need to be carried out weekly while construction debris is flushed out of a new loop.

M-04 Condenser fin pack. Isolate the fans, then wash the fin pack from the fan side outwards with water at no more than 3 bar nozzle pressure. Do not use pressure washers, acids or alkaline cleaners; they deform the fins and attack the coating. Straighten bent fins with a fin comb.

M-05 Leak inspection. Inspect all brazed joints, flare nuts, Schrader valves and the compressor terminal plate with an electronic detector rated for A2L refrigerants. Units whose total charge exceeds 10 kg must be leak-checked every three months; smaller charges are inspected every six months. Record the result and the detector serial number.

M-06 Glycol concentration. Draw a sample from the drain cock and measure with a refractometer calibrated for propylene glycol. Top up with premixed solution, never with plain water, if the reading has drifted below the commissioned value.

M-07 Power terminals. With the unit locked out and the drive discharged, re-torque the incoming supply terminals to 12 Nm and the compressor contactor terminals to 4 Nm. Look for discoloured insulation that would indicate an overheating joint.

M-08 Condenser fans. Spin each fan by hand with the unit isolated and listen for bearing roughness. Check blades for cracks and for balance weights that have come loose. Confirm that the fan guard grille is secure.

M-09 Flow switch test. With the compressor stopped, close the outlet isolating valve slowly while watching the display. FS1 should open and code E-4021 should be logged within about 10 seconds of flow falling below the minimum.

M-10 Water analysis. Send a sample of the process water to a laboratory and compare the result with the limits in Section 10.4. Keep the report with the logbook; it forms part of the water treatment record referred to in the warranty.

M-11 Drip tray. Remove the lower rear panel, lift out any debris from the drip tray and flush the drain line with a hose until water runs freely at the outlet.

M-12 Filter-drier. Replace the filter-drier cartridge PN-2208-F every 8,000 operating hours or whenever the refrigerant circuit is opened, whichever occurs first. A temperature drop of more than 2 K across the filter-drier shell during operation indicates a restriction and calls for earlier replacement.

M-13 Clock battery. Replace the CR2032 cell on the TCU-5 board. Parameters are held in non-volatile memory and are not lost, but the date and time must be re-entered after the battery change.

M-14 Annual inspection. The owner should arrange a comprehensive inspection by a Norvale-certified technician once every twelve months. The visit includes all twelve-month tasks, a review of trend data, a check of operating pressures against the commissioning record and a firmware update where one is available.

M-15 Oil acid test. Take an oil sample from the compressor service valve and test it with an acid test kit suitable for POE oil. An acid result calls for oil replacement, a new filter-drier and investigation of the cause, which is usually moisture or overheating.

M-16 Flow switch paddle. On units fitted with a paddle-type flow switch, replace the paddle and its seal. Units with a thermal dispersion switch need no periodic part replacement.

5.4 Water treatment record

The owner is responsible for keeping the process water within the quality limits in Section 10.4 and for maintaining a written water treatment log. The log should show the date, the inhibitor dose, the conductivity and pH readings and the name of the person carrying out the check. This record is requested in support of any warranty claim concerning the evaporator or the water-side components.

6 Sensors and Components

6.1 Sensor overview

ID	Measured quantity	Type	Part number
T1	Entering water temperature	Immersion NTC	PN-5110-T
T2	Compressor discharge temperature	Clamp-on PT1000	PN-5120-T
T3	Leaving water temperature	Immersion NTC	PN-5110-T
T4	Liquid line temperature	Clamp-on NTC	PN-5140-C
T5	Ambient air temperature	Shielded NTC	PN-5150-A
T6	Suction line temperature	Clamp-on NTC	PN-5160-C
P1	Discharge pressure	Transducer 0-50 bar	PN-5210-H
P2	Suction pressure	Transducer 0-17 bar	PN-5220-L
P3	Evaporator differential pressure	Transducer 0-2 bar	PN-5230-D
FS1	Evaporator flow	Paddle or thermal	PN-4450-F

On the X200-HD, sensors T2, T4, T6, P1 and P2 are duplicated for the second circuit and carry the suffix B on the display and harness labels (for example T2B). The part numbers are identical for both circuits.

6.2 Temperature sensors

T1 and T3 are immersion thermistors screwed into pockets in the evaporator inlet and outlet headers. Apply thermal paste when refitting them; a dry pocket introduces a lag that makes leaving water control unstable. The two sensors are interchangeable and share a part number.

T2 is a platinum resistance sensor clamped to the discharge pipe 150 mm from the compressor. It feeds the discharge temperature protection described under codes P-4201 and E-4201. The clamp must be tight and the sensor covered with the insulating sleeve supplied, or the reading will be several kelvin low.

T4 is a clamp-on NTC thermistor strapped to the liquid line between the filter-drier and the EEV. Its reading, together with the saturation temperature derived from P1, gives the liquid subcooling. T6 is a similar clamp-on sensor fitted to the suction pipe 200 mm upstream of the compressor, and is used with P2 for superheat control.

Subcooling is calculated continuously as the saturation temperature corresponding to the P1 reading minus the liquid temperature at T4. Healthy circuits at design conditions show between 5 and 8 K. Because a charge shortfall shrinks the column of liquid backing up in the condenser, a subcooling value below 3 K usually points to an undercharged circuit,

provided the condenser is clean and the fans are running at their commanded speed. Subcooling above 12 K suggests overcharge or a restriction downstream of the receiver.

T5 sits behind a radiation shield on the left-hand end panel. Direct sunlight on an unshielded sensor causes the fan control to over-react on bright days, so the shield must always be refitted after service.

6.3 Pressure transducers

P1 is a 0 to 50 bar ratiometric transducer on the discharge line. It provides the high-pressure cutout signal, the condensing temperature for fan control and the input to the subcooling calculation.

P2 is a 0 to 17 bar ratiometric transducer on the suction line. It is the basis of superheat control, and the controller trips the low-pressure cutout when suction falls below 1.8 bar for more than 30 seconds after the start-up bypass period. P2 also detects evaporator freezing risk on the X200 and X200-HD, where the saturated suction temperature is compared with the leaving water temperature.

P3 measures the pressure drop across the evaporator and is used by the controller to estimate water flow. It backs up FS1 and raises code E-4022 if the differential rises far above its commissioned value.

6.4 Flow switch

FS1 is mounted in the evaporator outlet pipe. Early production units carry a paddle-type switch; units built from week 30 of 2025 carry a thermal dispersion switch with no moving parts. Both types use part number PN-4450-F with a different suffix kit, and the controller must be told which is fitted through the FS.TYP parameter available from firmware v2.3.1.

6.5 Compressors

The X200 and X200-L use one variable-speed scroll compressor; the X200-HD uses two. Each compressor carries an internal motor protector embedded in the stator windings. The internal motor protector opens when winding temperature exceeds its rating, typically after prolonged operation at high condensing pressure combined with elevated suction superheat, and it will not reclose until the motor has cooled, which can take 45 to 90 minutes. Common contributors are a fouled condenser during high ambient conditions, a reduced refrigerant charge, which weakens the cooling effect of returning suction gas on the motor, and supply voltage imbalance above 2 percent. The controller reports this condition as E-2001, or E-2010 for the second compressor of an X200-HD.

Crankcase heaters of 70 W keep the oil free of dissolved refrigerant during standby. A compressor that has been without heater power for more than 4 hours must have the heater energised for 12 hours before it is started again.

6.6 Electronic expansion valve

The EEV (PN-3302-E) is a stepper-motor valve with 480 steps from closed to fully open. It is driven directly by the TCU-5 and closes fully on power loss using the energy stored in a small capacitor module. The superheat setpoint is 6 K on the X200 and X200-HD and 5 K on the X200-L.

6.7 Condenser fans

The condenser uses electronically commutated fans with integral speed controllers driven by a 0-10 V signal from the TCU-5. The condenser fan motor assembly for the X200-HD is supplied as PN-7731-B; standard X200 and X200-L units use PN-7731-A, which has a smaller motor frame and different mounting centres. The fan guard grille, PN-7713-B, is common to all variants.

Each fan reports its motor current to the controller over the fan bus. A marked difference between the phase currents of one motor usually indicates a winding fault or a loose connector at the fan terminal box, and is reported as E-4012.

6.8 Controller and display

The TCU-5 main board (PN-6605-C) carries the processor, the I/O circuits and the fan bus interface. The 7-inch display (PN-6610-K) connects to it over a shielded RS-485 cable up to 10 m long. A replacement board is supplied with blank parameters; the parameter set can be restored from the USB backup made at commissioning or from the Norvale service portal using the unit serial number.

6.9 Spare parts list

Part number	Description	Models
PN-2208-F	Filter-drier cartridge	All
PN-3302-E	Electronic expansion valve	All
PN-4410-S	Y-strainer 0.8 mm, DN65	X200, X200-L
PN-4411-S	Y-strainer 0.8 mm, DN80	X200-HD
PN-4450-F	Flow switch FS1	All
PN-5110-T	Water temperature sensor T1 / T3	All
PN-5120-T	Discharge temperature sensor T2	All
PN-5140-C	Liquid line sensor T4	All
PN-5150-A	Ambient sensor T5	All
PN-5160-C	Suction line sensor T6	All
PN-5210-H	Discharge pressure transducer P1	All

Part number	Description	Models
PN-5220-L	Suction pressure transducer P2	All
PN-5230-D	Differential pressure transducer P3	All
PN-6605-C	TCU-5 main board	All
PN-6610-K	7-inch touch display	All
PN-7713-B	Fan guard grille	All
PN-7731-A	Fan motor assembly	X200, X200-L
PN-7731-B	Fan motor assembly, heavy duty	X200-HD
PN-8820-G	Evaporator header gasket kit	All

7 Troubleshooting

7.1 Approach

Begin every fault investigation by reading the alarm history and the trend log. Many faults are the second or third symptom of an earlier event; a high discharge temperature trip, for example, is often preceded by hours of rising condensing pressure. Note the time stamps, then compare the conditions at the moment of the trip with the commissioning record.

Work from the simplest possible cause towards the most complex. Check supply voltage, water flow and the condition of the condenser before suspecting sensors, and check sensors before suspecting the controller. Replace the controller board only after its inputs have been verified with a calibrated meter.

7.2 Unit will not start

If the display is dark, check the control transformer fuses F1 and F2 and the 24 V supply at terminals X1:1 and X1:2. If the display is lit but the unit remains in standby, check whether the remote start contact is closed, whether a lockout alarm needs to be reset, and whether the display shows the anti-short-cycle countdown. A unit that stopped because of a supply failure will not restart by itself on older firmware; see Section 4.6.

Code E-1002 at every power-up indicates reversed phase rotation. Swap two incoming phases at the main isolator, never at the compressor or drive terminals.

7.3 Insufficient cooling

If the leaving water temperature stays above setpoint while the compressor runs at full speed, first confirm that the load has not grown beyond the capacity of the unit. Compare the temperature difference across the evaporator with the design value: a larger difference than design points to reduced flow; a smaller one, with the compressor at maximum, points to a refrigeration-side problem.

On the refrigeration side, examine suction pressure, superheat and subcooling together. Low suction pressure with high superheat and low subcooling indicates a shortage of charge. Low suction pressure with high superheat and normal or high subcooling indicates a restriction such as a blocked filter-drier or a sticking EEV. High suction pressure with low superheat at full speed suggests worn compressor scrolls.

7.4 High condensing pressure

High discharge pressure is most often caused by poor heat rejection at the condenser. Check that all fans run and reach their commanded speed, that the fin pack is not obstructed, that nothing has been stored against the air inlet, and that warm discharge air is not being drawn back in. The controller compares the condensing temperature with the

ambient temperature measured at T5; if the difference stays above 22 K it raises E-4210 as an early indication that the coil needs attention.

If the condenser is clean and the fans are healthy, consider non-condensable gases in the circuit or an overcharge. Both show as high subcooling together with high head pressure. Non-condensables normally enter during a repair that was not evacuated to below 500 microns.

7.5 Compressor overload trips

Repeated E-2001 events, particularly in the afternoon of hot days, usually trace back to the condenser. Clean the fin pack and check the fans before looking further. If the trips continue, measure the compressor supply voltage and the imbalance between phases under load, and check the refrigerant charge using the subcooling method described in Section 6.2.

Do not attempt to reset the motor protector by cooling the compressor shell with water. Rapid cooling of the shell can distort the scroll set and is excluded from warranty cover.

7.6 Flow alarms

A flow alarm at start-up on a new installation is nearly always caused by trapped air or a closed valve. A flow alarm that develops over weeks on an established installation is nearly always caused by debris collecting in the strainer. Where flow alarms occur only intermittently, look for an external pump that loses prime, or a flow switch paddle that has been bent by debris.

If the strainer is clean and flow has been confirmed with a portable flowmeter, check the FS.TYP setting. A thermal dispersion switch configured as a paddle type responds too slowly and may produce nuisance alarms during start-up.

7.7 Abnormal noise and vibration

A steady rattle from the pump that changes with the position of the discharge valve points to the pump suction conditions rather than to the chiller; refer to Section 3.5. A rumble from the refrigeration module that appears only at low compressor speed can be caused by a resonance in the pipework; the resonance skip bands can be adjusted under Service > Compressors > Skip Frequencies.

A rhythmic knocking from the condenser section is usually a fan blade touching the guard or a loose balance weight. Stop the fans and inspect as described for task M-08.

7.8 Water around the unit

Before searching for a leak in the process pipework, check the drip tray outlet described in Section 3.8. Condensate volume rises sharply in humid weather and at low leaving

water temperatures, and a partly blocked outlet that coped in winter may overflow in summer. Also check the automatic air vent, which can discharge water if its float sticks.

7.9 Communication problems

For building management communication, the TCU-5 supports Modbus RTU on terminals X6:1 (A), X6:2 (B) and X6:3 (reference). The default address is 1, at 19200 baud, 8 data bits, even parity and one stop bit. If the building management system reports a timeout, check termination resistors at both ends of the bus and confirm that no other device uses the same address.

8 Error Code Reference

8.1 Code structure

Codes beginning with E are faults. A fault stops the affected compressor or the whole unit and is either reset automatically when the condition clears (Auto) or must be acknowledged at the display (Manual). Three occurrences of an Auto fault within 60 minutes convert it to a Manual lockout. Codes beginning with P are advisory prompts: they do not stop the unit but indicate that attention is needed or that a special operating mode is active.

The first digit after the letter identifies the subsystem: 1 for supply and drive, 2 for compressors, 3 for sensors, 4 for the refrigerant and water circuits, and 5 for the controller and communications. Several codes differ only in the order of their digits or in their prefix letter. Always read the full code from the display, including the letter, before consulting the entries below.

8.2 Code entries

E-1001 Fault / Reset: Auto

Description: One phase of the incoming supply is missing.

Possible causes: Blown upstream fuse; loose supply terminal; failed supply contactor.

Action: Measure all three phase voltages at the main isolator. Restore the missing phase before resetting.

E-1002 Fault / Reset: Manual

Description: Incoming phase sequence is reversed.

Possible causes: Supply cable connected with incorrect rotation after installation or grid work.

Action: Swap two phases at the main isolator only.

E-1010 Fault / Reset: Auto

Description: Supply voltage has been below 360 V for more than 5 seconds.

Possible causes: Weak supply; undersized cable; heavy starting loads elsewhere on the site.

Action: Log the supply voltage over a working day. Discuss persistent undervoltage with the site electrician.

E-1011 Fault / Reset: Auto

Description: Supply voltage is above 440 V.

Possible causes: Transformer tap setting; light-load conditions on the site network.

Action: Confirm the reading with a meter. The unit restarts automatically when voltage returns within limits.

E-1020 Fault / Reset: Auto

Description: DC link overvoltage in the compressor drive.

Possible causes: Supply transients; rapid deceleration of the compressor.

Action: If repeated, fit the line reactor kit and check the drive deceleration ramp parameter.

E-1021 Fault / Reset: Auto

Description: Compressor drive heatsink over temperature.

Possible causes: Blocked drive cooling fan; panel filter clogged; ambient above rating.

Action: Clean the panel filter and check the drive fan runs when the compressor is operating.

E-2001 Fault / Reset: Manual

Description: Compressor 1 motor protector is open.

Possible causes: Thermal overload of the motor windings; see Section 6.5 and Section 7.5.

Action: Allow the motor to cool fully. Investigate condenser condition, charge and supply imbalance before restarting.

E-2002 Fault / Reset: Manual

Description: Compressor failed to start after three attempts.

Possible causes: Drive fault; seized compressor; open motor winding.

Action: Measure winding resistances at the compressor terminal plate and compare them with each other.

E-2010 Fault / Reset: Manual

Description: Compressor 2 motor protector is open (X200-HD only).

Possible causes: As for E-2001, applied to the second circuit.

Action: As for E-2001. The first circuit continues to run alone while this fault is active.

E-2105 Fault / Reset: Auto

Description: Maximum number of compressor starts per hour exceeded.

Possible causes: Load too small for the unit; insufficient loop volume; setpoint band too narrow.

Action: Review the trend log and consider a buffer tank as described in Section 4.4.

E-2110 Fault / Reset: Auto

Description: Oil return cycle did not complete.

Possible causes: Prolonged operation at minimum speed; EEV not opening during the oil return sequence.

Action: Check EEV operation in manual mode and confirm that the oil return interval parameter is at its default.

E-3101 Fault / Reset: Auto

Description: Entering water sensor T1 reads open or short circuit.

Possible causes: Damaged cable; water in the connector; failed thermistor.

Action: Measure sensor resistance and compare with the table on the sensor label. Replace with PN-5110-T.

E-3102 Fault / Reset: Manual

Description: Discharge temperature sensor T2 reads outside its valid range.

Possible causes: Sensor detached from pipe; broken lead; failed element.

Action: Refit or replace with PN-5120-T. The compressor is locked out while T2 is invalid.

E-3103 Fault / Reset: Manual

Description: Leaving water sensor T3 reads open or short circuit.

Possible causes: Damaged cable; failed thermistor.

Action: Replace with PN-5110-T. The unit cannot control without a valid T3 reading.

E-3104 Fault / Reset: Auto

Description: Liquid line sensor T4 reads outside its valid range.

Possible causes: Loose clamp; damaged lead.

Action: Refit or replace with PN-5140-C. Subcooling values are not available while this fault is active.

E-3105 Fault / Reset: Auto

Description: The ambient air temperature sensor T5 reads open or short circuit.

Possible causes: Cable damaged by weather or rodents; failed sensor.

Action: Replace with PN-5150-A. Fan control falls back to a fixed head pressure target meanwhile.

E-3106 Fault / Reset: Auto

Description: Suction line sensor T6 reads outside its valid range.

Possible causes: Loose clamp; damaged lead; missing insulation.

Action: Refit or replace with PN-5160-C. The EEV moves to a fixed safe opening while T6 is invalid.

E-3201 Fault / Reset: Manual

Description: Discharge pressure transducer P1 signal out of range.

Possible causes: Connector corrosion; failed transducer; 5 V supply fault.

Action: Check the 5 V reference at the connector, then replace with PN-5210-H if the supply is correct.

E-3202 Fault / Reset: Manual

Description: Suction pressure transducer P2 signal out of range.

Possible causes: Connector corrosion; failed transducer.

Action: Replace with PN-5220-L after recovering refrigerant if the transducer has no Schrader isolation.

E-3203 Fault / Reset: Auto

Description: Differential pressure transducer P3 signal out of range.

Possible causes: Impulse line blocked; failed transducer.

Action: Blow through the impulse lines and verify the transducer output against a test gauge.

E-4001 Fault / Reset: Manual

Description: Discharge pressure exceeded the electronic high-pressure cutout at 41.5 bar.

Possible causes: Condenser fouling; fan failure; overcharge; non-condensable gases.

Action: Follow Section 7.4. Do not reset more than twice without finding the cause.

E-4002 Fault / Reset: Auto

Description: Suction pressure remained under its lower limit after the start bypass.

Possible causes: Shortage of charge; blocked filter-drier; EEV stuck closed; very low water flow.

Action: Check superheat and subcooling as described in Section 7.3.

E-4012 Fault / Reset: Auto

Description: Condenser fan motor current imbalance between phases exceeds 18 percent.

Possible causes: Winding fault in the fan motor; loose connector at the fan terminal box.

Action: Measure phase currents at the fan. Replace the fan motor assembly with the part number for the model variant.

E-4021 Fault / Reset: Auto

Description: Evaporator flow lost: FS1 has been open for more than 10 seconds while flow was demanded.

Possible causes: Blocked strainer; closed valve; pump failure; air in the loop.

Action: Confirm the pump is running, then first inspect the inlet strainer PN-4410-S for trapped debris.

E-4022 Fault / Reset: Auto

Description: Evaporator differential pressure far above its commissioned value.

Possible causes: Partly closed valve; scaled evaporator plates; frozen evaporator.

Action: Compare the P3 value with the commissioning record and check water quality history.

E-4102 Fault / Reset: Auto

Description: Subcooling has remained under 2 K for 15 minutes at full load.

Possible causes: Shortage of refrigerant charge; T4 sensor displaced.

Action: Leak-test the circuit before adding refrigerant. Never top up without locating the leak.

E-4110 Fault / Reset: Auto

Description: Suction superheat has remained above 15 K with the EEV fully open.

Possible causes: Shortage of charge; restriction upstream of the EEV; failed EEV coil.

Action: Check the EEV coil resistance and the temperature drop across the filter-drier.

E-4120 Fault / Reset: Manual

Description: Freeze protection: leaving water below 3 °C on X200 and X200-HD, or below the glycol-dependent limit on X200-L.

Possible causes: Low flow; incorrect setpoint; incorrect glycol percentage entered.

Action: Check flow and the entered glycol value before resetting. Inspect the evaporator for frost damage.

E-4201 Fault / Reset: Manual

Description: Compressor discharge temperature at T2 has exceeded the 118 °C trip limit.

Possible causes: High compression ratio; high suction superheat; shortage of charge.

Action: Allow the compressor to cool and investigate superheat and condensing pressure before restarting.

E-4210 Fault / Reset: Auto

Description: Condensing temperature has stayed more than 22 K above ambient.

Possible causes: Obstructed fin pack; recirculation of discharge air; fan fault.

Action: Carry out task M-04 and check airflow clearances.

E-4301 Fault / Reset: Auto

Description: Drip tray level switch has operated (optional kit).

Possible causes: Obstructed condensate drain.

Action: Carry out task M-11.

E-5001 Fault / Reset: Manual

Description: Controller parameter memory checksum error.

Possible causes: Interrupted firmware update; board fault.

Action: Reload parameters from the commissioning backup. Replace PN-6605-C if the error returns.

E-5010 Fault / Reset: Auto

Description: Modbus communication timeout with the building management system.

Possible causes: Wiring fault; address conflict; termination missing.

Action: See Section 7.9. The unit continues to run on its local setpoint.

E-5020 Fault / Reset: Auto

Description: Communication with the display lost.

Possible causes: Damaged RS-485 cable; display failure.

Action: Check the cable and connectors between PN-6605-C and PN-6610-K.

E-5101 Fault / Reset: Auto

Description: Controller clock battery voltage is low.

Possible causes: Battery at end of life.

Action: Carry out task M-13.

P-1001 Advisory / Reset: None

Description: The programmed service interval has elapsed.

Possible causes: Normal service reminder.

Action: Carry out the scheduled tasks and reset the counter from the Service menu.

P-2105 Advisory / Reset: None

Description: Compressor starts per hour are approaching the permitted maximum.

Possible causes: Short cycling due to low load.

Action: Review the loop volume. This prompt precedes E-2105.

P-4012 Advisory / Reset: None

Description: Night setback is active; fan speed is capped.

Possible causes: Normal operation within the programmed night setback window.

Action: No action required. See Section 4.7.

P-4021 Advisory / Reset: None

Description: The flow switch type parameter has not been set after a firmware update.

Possible causes: Unit updated from firmware v2.2.x or v2.3.0 to v2.3.1 without completing the new parameter.

Action: Set FS.TYP to PADDLE or THERMAL under Settings > Water > Flow Switch.

P-4201 Advisory / Reset: None

Description: Discharge temperature at T2 is above 110 °C; compressor speed is being reduced.

Possible causes: Conditions approaching the E-4201 trip limit.

Action: Check condenser condition and superheat before the unit reaches a trip.

8.3 Resetting faults

Manual faults are reset from the Alarms screen with a service-level password once the cause has been corrected. Resetting a fault without correcting its cause is recorded in the alarm history with the password level used. Repeated resets of E-4001 or E-4201 without corrective action are treated as misuse for warranty purposes.

9 Warranty and Service Policy

9.1 Scope

The clauses in this section reproduce the Norvale Standard Warranty Terms, document NSWT-2025, as they apply to AquaTherm X200 series products. Clause numbers follow that document and are quoted by those numbers in all warranty correspondence. Where a supply contract contains different terms, the contract prevails.

9.2 Warranty terms

Clause 7.1 The standard warranty period is 24 months from the date of commissioning or 30 months from the date of dispatch from the factory, whichever expires first. Commissioning is evidenced by the returned form NTS-CF-200.

Clause 7.2 The compressor warranty is extended to 60 months from commissioning provided the unit is registered on the Norvale service portal within 90 days of commissioning and the annual inspection (task M-14) is carried out and recorded each year.

Clause 7.3 Electronic parts are covered for a fixed term regardless of registration: sensors, transducers and controller boards are warranted for 24 months from commissioning.

Clause 7.4 Where a compressor is replaced under warranty, the failed compressor must be returned to Norvale within 30 calendar days of the replacement being dispatched, with the failure report attached. A compressor not returned within this period is invoiced at the current replacement rate.

Clause 7.5 Replacement refrigerant is covered only where the loss results from a manufacturing defect. Refrigerant lost through damage in service, incorrect repair or vandalism is not covered.

Clause 7.6 The warranty does not extend to damage attributable to scale, fouling, corrosion or inadequate water treatment, including failures of the evaporator where the water treatment record described in Section 5.4 has not been kept.

Clause 7.7 The warranty is void if the controller firmware has been modified or replaced by anyone other than Norvale or an authorised service partner, or if parts not supplied by Norvale have been fitted to the refrigerant circuit.

Clause 7.8 Consumable items are excluded, including strainer screens, filter-drier cartridges, gaskets, flow switch paddles and the controller clock battery.

9.3 Service terms

Clause 8.1 Norvale or its authorised partner will attend a warranty breakdown within 48 hours of the call being logged, on working days. Response times for sites more than 150 km from the nearest service base are agreed individually.

Clause 8.2 Attendance outside normal working hours, at weekends or on public holidays is available on request and is billed at 1.5 times the standard labour rate, even where the parts are covered by warranty.

Clause 8.3 Travel to sites more than 150 km from the nearest service base is charged at cost for the distance in excess of 150 km.

Clause 8.4 Owners of X200-HD units serving critical processes may enter a standby agreement under which a temporary chiller is provided while a major repair is carried out.

9.4 Making a claim

Warranty claims are submitted through the Norvale service portal. Each claim must include the unit serial number, the alarm history export from the controller, a copy of the relevant logbook pages, photographs of the failed part and its identification label, and a description of the work carried out. Claims submitted without the alarm history export are returned for completion.

Parts replaced under warranty become the property of Norvale. Unless the claim acknowledgement states otherwise, the failed part must be kept for 60 days after the claim is submitted in case it is requested for inspection. This general retention period does not replace the specific compressor return requirement in the warranty terms above.

9.5 Registration

Registering the unit on the service portal gives access to the extended compressor warranty, firmware downloads, parameter backups and the full technical bulletin archive. Registration requires the serial number and the commissioning date and can be completed by the installer on behalf of the owner.

10 Technical Specifications

10.1 Performance

Parameter	X200	X200-L	X200-HD
Nominal cooling capacity	62 kW	44 kW at -5 °C	88 kW
Leaving fluid range	5 to 20 °C	-8 to 15 °C	5 to 20 °C
Ambient operating range	-15 to 46 °C	-15 to 46 °C	-15 to 50 °C
Nominal water flow	2.95 L/s	2.40 L/s	4.05 L/s
Minimum water flow	1.6 L/s	1.4 L/s	2.2 L/s
Evaporator pressure drop	38 kPa	46 kPa	41 kPa
Compressors / circuits	1 / 1	1 / 1	2 / 2
Refrigerant	R-454B	R-454B	R-454B
Refrigerant charge	6.8 kg	7.4 kg	2 x 7.1 kg

10.2 Electrical

Parameter	X200	X200-L	X200-HD
Supply	400 V 3~ 50 Hz	400 V 3~ 50 Hz	400 V 3~ 50 Hz
Minimum circuit ampacity	58 A	54 A	82 A
Maximum fuse rating	80 A	80 A	110 A
Nominal power input	19.4 kW	21.0 kW	27.8 kW
Control voltage	24 V AC	24 V AC	24 V AC

10.3 Physical

Parameter	X200	X200-L	X200-HD
Length x width x height	2150 x 1100 x 1850 mm	2150 x 1100 x 1850 mm	2900 x 1100 x 1950 mm
Operating weight	612 kg	655 kg	948 kg
Water connections	DN65 grooved	DN65 grooved	DN80 grooved
Maximum water-side pressure	6.5 bar	6.5 bar	8.0 bar
Condenser fans	2	2	3

10.4 Water quality limits

Parameter	Limit
pH	7.5 to 9.0
Electrical conductivity	10 to 500 µS/cm
Chlorides	below 50 mg/l
Total hardness	4 to 8.5 °dH
Free chlorine	below 0.5 mg/l
Suspended solids	below 10 mg/l, particle size below 0.8 mm

10.5 Notes on specifications

Capacities are stated at 12/7 °C water and 35 °C ambient air for the X200 and X200-HD, and at 0/-5 °C with 35 percent propylene glycol for the X200-L. Capacity falls by approximately 3 percent for every 1 K rise in ambient above 35 °C and by approximately 3.5 percent for every 1 K fall in leaving temperature below the rating point.

The X200-HD carries a total refrigerant charge of 14.2 kg divided equally between its two circuits. The reinforced headers and heavier evaporator plates of this variant mean it is rated for a maximum water inlet pressure of 8.0 bar, which allows direct connection to most high-rise building loops without a break tank.

At altitudes above 1,000 m, reduce the stated capacity by 1 percent per additional 300 m, and check with Norvale whether the fan speed limits need to be adjusted. Performance figures apply to units with clean heat exchangers and fluid within the quality limits above.

The design life of the refrigeration circuit is 15 years when maintained according to Section 5. The compressor is designed for 60,000 operating hours under rated conditions.

11 Glossary

A2L: A refrigerant safety class for substances of low toxicity and low flammability.

Anti-short-cycle timer: A controller timer that prevents the compressor from being restarted too soon after it last stopped.

Approach: The temperature difference between a refrigerant saturation temperature and the fluid or air it exchanges heat with.

Buffer tank: A storage tank added to the loop to increase water volume and stabilise control.

Condenser fouling: Accumulation of deposits on the air side of the condenser that impairs heat rejection.

Crankcase heater: An electric heater that keeps compressor oil warm to limit refrigerant migration during standby.

Cutout: A protective device or function that stops the compressor when a limit is reached.

DC link: The intermediate direct-current stage of a variable-speed drive, containing large capacitors.

Differential pressure: The difference between two pressures, here across the evaporator water side.

EC fan: A fan driven by an electronically commutated motor with integral speed control.

EEV: Electronic expansion valve; a stepper-motor valve metering refrigerant into the evaporator.

Filter-drier: A replaceable cartridge that removes moisture and particles from the liquid refrigerant.

FS.TYP: Controller parameter that identifies the type of flow switch fitted.

Glycol: An additive that lowers the freezing point of the process fluid.

Head pressure: An informal term for the condensing or discharge pressure.

Lead-lag: An arrangement in which the duty of starting first is shared between two compressors.

Lockout: A fault state that requires manual acknowledgement before the unit may restart.

Motor protector: A temperature-sensitive switch inside the compressor motor that interrupts operation on overheating.

Non-condensables: Gases such as air or nitrogen trapped in a refrigerant circuit that do not condense at operating conditions.

NTC: Negative temperature coefficient thermistor; its resistance falls as temperature rises.

POE oil: Polyolester lubricant used with R-454B compressors; strongly hygroscopic.

Refrigerant subcooling: The amount by which liquid refrigerant is cooled below its saturation temperature.

Saturation temperature: The boiling or condensing temperature of a refrigerant at a given pressure.

Scroll compressor: A compressor using one fixed and one orbiting spiral to compress refrigerant vapour.

Short cycling: Frequent starting and stopping of a compressor caused by low load or insufficient system volume.

Superheat: The amount by which refrigerant vapour is heated above its saturation temperature.

Thermal dispersion switch: A flow switch that detects flow by the cooling effect of the fluid on a heated probe.

Thermal overload: Overheating of a motor winding beyond its rated temperature, leading to protective shutdown.

Trend log: The controller record of temperatures and pressures sampled at fixed intervals.

Undercharge: A refrigerant charge smaller than the quantity specified for the circuit.

Vapour-compression cycle: The refrigeration process of evaporation, compression, condensation and expansion.